

HAT-P-18b

R-2395

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_220705 · created 2026-08-21T00:04:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459765.88593 +/- 0.00045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.053
Transit depth (Rp/Rs)^2	3.33 +/- 1.93 [%]
Orbital Inclination (inc)	89.77 +/- 1.85
Airmass coefficient 1 (a1)	1.018 +/- 0.039
Airmass coefficient 2 (a2)	-0.018 +/- 0.031
Scatter in the residuals of the lightcurve fit is	3.9 %
Best Comparison Star	#1 - [250, 384]
Optimal Aperture	3.69
Optimal Annulus	10.34
Transit Duration (day)	0.094 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.182 $\pm$ 0.053 vs 0.1356 (deviation 0.88 σ)
Duration fit vs published	0.094 d vs 0.1129 d (deviation 1.11 σ)
Mid-time O–C	0.3 min
Depth SNR	3.4
Residual scatter	3.9 %

Light curve

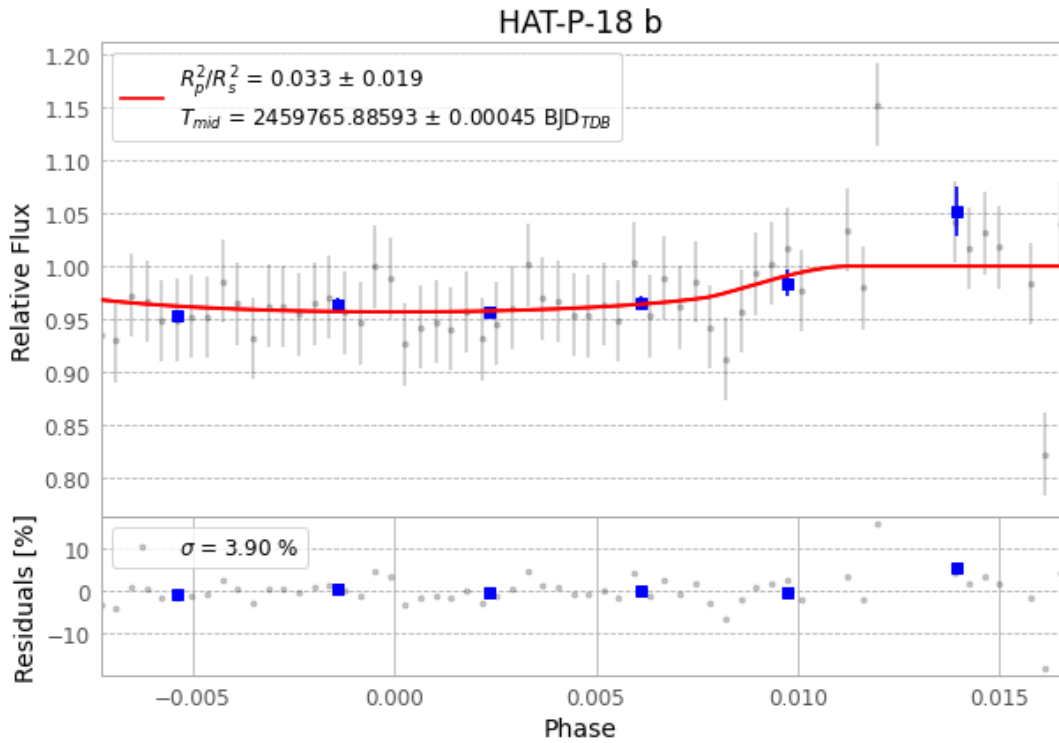


Figure 1 — FinalLightCurve_HAT-P-18 b_2022-07-05.png (SHA-256 ad7e0974c4cbcc7a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [272, 302], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e957896bc5b0e1cd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), HATP-18220705081213.FITS → HATP-18220705112112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-220705.log (69ed6fa82775...), FinalLightCurve_HAT-P-18 b_2022-07-05.png (ad7e0974c4cb...), FinalLightCurve_HAT-P-18 b_2022-07-05.csv (e1abe6149897...)

Compute resources

Wall time 405.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:04Z.